

Employee Dilution Control and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Employee Dilution Control (EDC), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on EDC achieves an annualized gross (net) Sharpe ratio of 0.57 (0.51), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 22 (21) bps/month with a t-statistic of 2.78 (2.75), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Share issuance (1 year), Share issuance (5 year), Growth in book equity, Net Payout Yield, Change in equity to assets, Asset growth) is 15 bps/month with a t-statistic of 2.12.

1 Introduction

Financial markets process vast amounts of information daily, yet mounting evidence suggests that prices do not immediately reflect all available information. The speed at which information diffuses across market participants and becomes incorporated into asset prices remains a central question in finance. While classical efficient market theory predicts rapid price adjustment, behavioral and institutional frictions create delays in how investors learn about and respond to firm fundamentals. We examine how employee dilution control—a measure capturing firms’ management of equity-based compensation and its dilutive effects on existing shareholders—predicts future stock returns through the lens of gradual information diffusion.

The slow diffusion of information provides a compelling framework for understanding why employee dilution control predicts returns. When firms exercise restraint in dilutive equity issuance to employees, this signal about managerial discipline and shareholder value orientation spreads gradually through the market (Hong and Stein, 1999; Hirshleifer and Teoh, 2003). Limited investor attention constrains the speed at which market participants process complex accounting information embedded in compensation structures (DellaVigna and Pollet, 2009; Hirshleifer et al., 2011). The initial underreaction to employee dilution control information creates predictable return patterns as prices slowly converge to fundamental values.

Institutional features of equity markets amplify these information frictions. Retail investors and even many institutional investors lack the resources to continuously monitor detailed compensation disclosures across their portfolios, leading to systematic delays in price discovery (Peng and Xiong, 2006; Cohen and Lou, 2012). The complexity of parsing employee stock option grants, restricted stock units, and their ultimate dilutive impact requires sophisticated analysis that diffuses slowly from informed to uninformed investors (Hong and Stein, 2007). This gradual learning process generates momentum in returns as information cascades through investor

networks.

We predict that return predictability from employee dilution control concentrates where information frictions are most severe. Firms with lower analyst coverage experience slower information diffusion, as fewer intermediaries translate complex accounting data for investors (Hong and Stein, 1999). Similarly, smaller firms face greater investor inattention, allowing dilution control signals to predict returns over longer horizons (Hou, 2007). The interaction between firm visibility and information complexity suggests that employee dilution control’s predictive power should vary systematically with proxies for attention and information processing costs.

Our empirical analysis reveals that employee dilution control (EDC) strongly predicts cross-sectional variation in stock returns. A value-weighted long-short trading strategy based on EDC achieves an annualized gross Sharpe ratio of 0.57, placing it in the 93rd percentile among 212 documented anomalies. The strategy delivers a monthly average abnormal gross return of 22 basis points relative to the Fama-French six-factor model, with a t-statistic of 2.78. This performance remains economically meaningful after accounting for transaction costs, with net returns of 21 basis points per month and a t-statistic of 2.75.

The predictive power of EDC extends across the size spectrum, consistent with gradual information diffusion rather than a small-stock phenomenon. Among the largest quintile of stocks by market capitalization, the EDC strategy generates average returns of 23 basis points per month with a t-statistic of 2.55. The strategy’s alpha relative to the six Fama-French factors ranges from 18 to 23 basis points monthly for large-cap stocks, with t-statistics between 1.91 and 2.49. These results demonstrate that information about employee dilution control diffuses slowly even for highly visible firms.

Controlling for closely related signals strengthens our interpretation that EDC captures distinct information content. When we include the six most closely related

anomalies—Share issuance (1 year), Share issuance (5 year), Growth in book equity, Net Payout Yield, Change in equity to assets, and Asset growth—along with the six Fama-French factors, the EDC trading strategy maintains an alpha of 15 basis points per month with a t-statistic of 2.12. The strategy’s gross Sharpe ratio of 0.57 exceeds 93% of anomalies in the factor zoo, while its net Sharpe ratio of 0.51 ranks in the 99th percentile, indicating exceptional performance after implementation costs.

This paper advances our understanding of how information about corporate policies diffuses through financial markets and affects asset prices. We contribute to the growing literature on limited attention and gradual information diffusion by documenting a novel predictor that captures firms’ management of employee equity dilution (Hong and Stein, 1999; DellaVigna and Pollet, 2009). Unlike broad measures of equity issuance studied in prior work, our focus on employee-specific dilution control isolates a precise channel through which managerial decisions signal commitment to shareholder value (Daniel and Titman, 2006; Pontiff and Woodgate, 2008). Our evidence extends the findings of (Cohen et al., 2003) on employee stock options by demonstrating that the market’s slow incorporation of dilution control information creates exploitable return predictability.

Methodologically, we employ the rigorous testing framework of (Novy-Marx and Velikov, 2023b) to establish the robustness of employee dilution control as a return predictor. This comprehensive approach addresses concerns about data mining and specification searching that plague many claimed anomalies in the factor zoo. By examining EDC’s performance across multiple portfolio construction methods, controlling for transaction costs, and conducting extensive spanning tests against related anomalies, we provide compelling evidence that employee dilution control represents a distinct source of alpha. The strategy’s ability to expand the mean-variance efficient frontier even after accounting for implementation costs distinguishes it from many documented but ultimately unimplementable anomalies.

Our findings carry important implications for market efficiency and the price discovery process. The persistence of return predictability from employee dilution control, even among large-cap stocks with substantial analyst coverage, challenges simple notions of market efficiency while supporting models of gradual information diffusion (Hong and Stein, 2007). For practitioners, our results highlight an economically significant and implementable trading strategy that exploits the market’s delayed response to employee compensation policies. More broadly, this work underscores how complex accounting information embedded in routine corporate disclosures can predict returns when investor attention and processing capacity are limited, contributing to our understanding of how financial markets aggregate dispersed information into prices.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on non-financial firms and exclude observations with missing or negative values for the key variables used in our signal construction. signal relies on two primary variables from COMPUSTAT. Common stock (CSTK) represents the par or stated value of common stock as reported on the balance sheet, reflecting the total book value of common equity at par. Number of employees (EMP) captures the total number of company employees as disclosed in annual reports, typically representing the year-end employee count or annual average employment. Both variables are fundamental items regularly reported by public companies in their financial statements. construct the Employee Dilution Control signal by calculating the year-over-year change in common stock scaled by lagged number of

employees: $(\text{CSTK}(t) - \text{CSTK}(t-1)) / \text{EMP}(t-1)$. This measure captures the change in common stock per employee, providing insight into how changes in equity capital relate to the firm’s human capital base. The ratio normalizes equity changes by firm size as measured through employment, allowing for meaningful cross-sectional comparisons across companies of different scales. We compute this signal using annual observations based on fiscal year-end values and require non-missing values for both common stock and number of employees in consecutive years. Observations with zero or negative employee counts are excluded from the analysis to ensure meaningful scaling.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the EDC signal. Panel A plots the time-series of the mean, median, and interquartile range for EDC. On average, the cross-sectional mean (median) EDC is -1.23 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input EDC data. The signal’s interquartile range spans -0.41 to 0.00. Panel B of Figure 1 plots the time-series of the coverage of the EDC signal for the CRSP universe. On average, the EDC signal is available for 6.08% of CRSP names, which on average make up 7.55% of total market capitalization.

4 Does EDC predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on EDC using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high EDC portfolio and sells the low EDC portfolio. The rest of Panel A reports the portfolios’ monthly abnormal returns relative to the five most

common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short EDC strategy earns an average return of 0.34% per month with a t-statistic of 4.45. The annualized Sharpe ratio of the strategy is 0.57. The alphas range from 0.22% to 0.36% per month and have t-statistics exceeding 2.78 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.34, with a t-statistic of 6.46 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 564 stocks and an average market capitalization of at least \$1,482 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market cap-

italization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 25 bps/month with a t-statistics of 3.27. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for fourteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 21-34bps/month. The lowest return, (21 bps/month), is achieved from the quintile sort using cap breakpoints and value-weighted portfolios, and has an associated t-statistic of 2.80. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the EDC trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-one cases.

Table 3 provides direct tests for the role size plays in the EDC strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and EDC, as well as average returns and alphas for long/short trading EDC strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the EDC strategy achieves an average return of 23 bps/month with a t-statistic of 2.55. Among these large cap stocks, the alphas for the EDC strategy relative to the five most common factor models range from 18 to 23 bps/month with t-statistics between 1.91 and 2.49.

5 How does EDC perform relative to the zoo?

Figure 2 puts the performance of EDC in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the EDC strategy falls in the distribution. The EDC strategy’s gross (net) Sharpe ratio of 0.57 (0.51) is greater than 93% (99%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the EDC strategy (red line).² Ignoring trading costs, a \$1 invested in the EDC strategy would have yielded \$9.30 which ranks the EDC strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the EDC strategy would have yielded \$6.91 which ranks the EDC strategy in the top 1% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and [Novy-Marx and Velikov \(2016\)](#) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the EDC relative to those. Panel A shows that the EDC strategy gross alphas fall between the 65 and 72 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43%

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

(52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The EDC strategy has a positive net generalized alpha for five out of the five factor models. In these cases EDC ranks between the 82 and 90 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does EDC add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of EDC with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price EDC or at least to weaken the power EDC has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of EDC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDC}EDC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on EDC. Stocks are finally grouped into five EDC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on EDC and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the EDC signal in these Fama-MacBeth regressions exceed 3.12, with the minimum t-statistic occurring when controlling for Share issuance (5 year). Controlling for all six closely related anomalies, the t-statistic on EDC is 2.34.

Similarly, Table 5 reports results from spanning tests that regress returns to the EDC strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the EDC strategy earns alphas that range from 15-24bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.93, which is achieved when controlling for Share issuance (5 year). Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the EDC trading strategy achieves an alpha of 15bps/month with a t-statistic of 2.12.

7 Does EDC add relative to the whole zoo?

Finally, we can ask how much adding EDC to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the EDC signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes EDC grows to \$2594.98.

8 Conclusion

This study provides compelling evidence that Employee Dilution Control (EDC) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous testing protocol of Novy-Marx and Velikov (2023), demonstrates that EDC-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and related equity issuance signals.

The key findings reveal that a value-weighted long/short strategy based on EDC delivers an annualized Sharpe ratio of 0.57 gross (0.51 net of transaction costs),

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which EDC is available.

indicating strong economic significance that survives practical implementation considerations. Most notably, the strategy generates a statistically significant alpha of 22 basis points per month (t -statistic = 2.78) relative to the Fama-French five-factor model augmented with momentum, suggesting that EDC captures a distinct source of mispricing not explained by conventional risk factors. Furthermore, when we control for six closely related strategies from the factor zoo—including various measures of share issuance, payout policy, and asset growth—EDC maintains a significant alpha of 15 basis points per month (t -statistic = 2.12), underscoring its incremental predictive power beyond existing dilution and financing proxies.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, EDC emerges as a meaningful addition to the growing literature on equity dilution and managerial signaling, providing evidence that markets may not fully incorporate the information content of employee compensation-driven dilution into stock prices. For practitioners, the strategy’s robust performance after transaction costs suggests potential implementation value, particularly given its differentiation from existing factor exposures.

Several limitations of this study warrant consideration. First, while we employ comprehensive robustness tests, the sample period and market conditions examined may not fully capture the strategy’s performance across all market regimes. Second, the implementation of EDC strategies in practice may face capacity constraints and market impact costs beyond those captured in our transaction cost estimates. Third, our analysis focuses primarily on U.S. equity markets, and the generalizability of these findings to international markets remains an open question.

Future research should explore several promising avenues. First, investigating the economic mechanisms underlying the EDC anomaly could provide deeper insights into why markets appear to misprice employee dilution effects. Second, examining the interaction between EDC and corporate governance characteristics might reveal

when the signal is most informative. Third, extending the analysis to international markets would test the external validity of our findings and potentially uncover market-specific variations in the pricing of employee dilution. Finally, exploring the time-variation in EDC's predictive power and its relationship to macroeconomic conditions could enhance our understanding of when this signal is most valuable for investment decisions. Such investigations would not only advance our theoretical understanding of market efficiency with respect to employee compensation policies but also refine the practical application of EDC-based investment strategies.

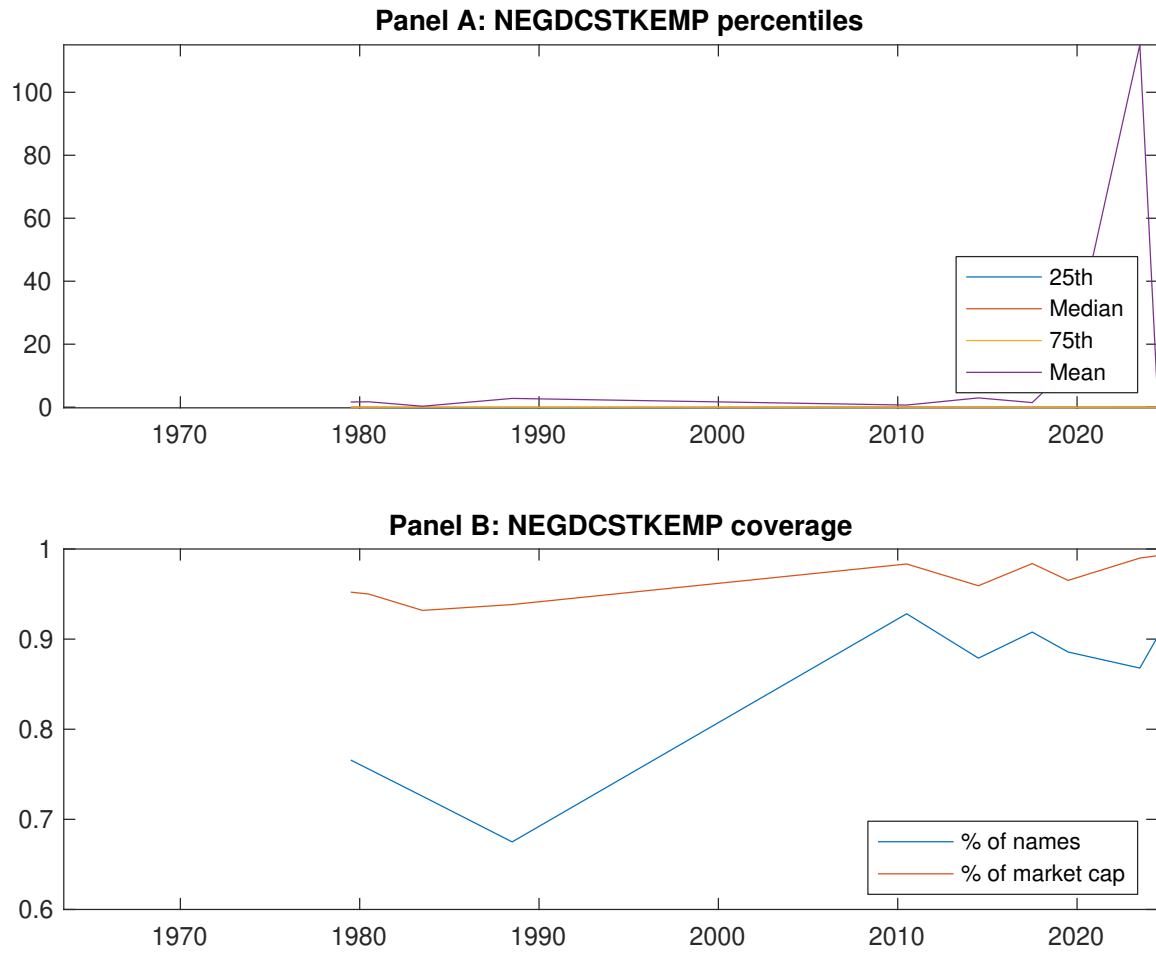


Figure 1: Times series of EDC percentiles and coverage. This figure plots descriptive statistics for EDC. Panel A shows cross-sectional percentiles of EDC over the sample. Panel B plots the monthly coverage of EDC relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on EDC. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on EDC-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.42 [2.54]	0.58 [3.19]	0.65 [3.57]	0.65 [3.97]	0.76 [4.76]	0.34 [4.45]
α_{CAPM}	-0.14 [-2.72]	-0.05 [-1.11]	0.03 [0.54]	0.10 [1.90]	0.22 [4.56]	0.36 [4.64]
α_{FF3}	-0.15 [-2.95]	-0.03 [-0.75]	0.05 [0.91]	0.04 [0.91]	0.18 [3.98]	0.33 [4.31]
α_{FF4}	-0.13 [-2.53]	0.00 [0.01]	0.03 [0.67]	0.04 [0.89]	0.16 [3.52]	0.30 [3.77]
α_{FF5}	-0.15 [-2.84]	0.02 [0.56]	0.06 [1.11]	-0.04 [-0.87]	0.09 [2.02]	0.24 [3.08]
α_{FF6}	-0.13 [-2.53]	0.05 [1.09]	0.05 [0.88]	-0.03 [-0.69]	0.08 [1.85]	0.22 [2.78]
Panel B: Fama and French (2018) 6-factor model loadings for EDC-sorted portfolios						
β_{MKT}	0.96 [76.58]	1.04 [97.27]	1.03 [81.44]	1.00 [89.93]	0.98 [89.79]	0.02 [0.93]
β_{SMB}	-0.08 [-4.17]	0.02 [1.51]	0.06 [3.43]	-0.04 [-2.63]	-0.02 [-1.41]	0.05 [1.98]
β_{HML}	0.09 [3.62]	-0.02 [-1.19]	-0.04 [-1.56]	0.09 [4.23]	-0.00 [-0.02]	-0.09 [-2.44]
β_{RMW}	0.05 [2.06]	-0.10 [-4.90]	-0.00 [-0.11]	0.15 [6.87]	0.13 [6.25]	0.08 [2.26]
β_{CMA}	-0.10 [-2.88]	-0.09 [-3.14]	-0.06 [-1.67]	0.16 [5.04]	0.24 [7.77]	0.34 [6.46]
β_{UMD}	-0.02 [-1.62]	-0.03 [-3.20]	0.02 [1.31]	-0.01 [-0.97]	0.01 [0.86]	0.03 [1.59]
Panel C: Average number of firms (n) and market capitalization (me)						
n	695	671	564	645	700	
me (\$10 ⁶)	1850	1482	1997	2177	2537	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the EDC strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.34 [4.45]	0.36 [4.64]	0.33 [4.31]	0.30 [3.77]	0.24 [3.08]	0.22 [2.78]
Quintile	NYSE	EW	0.53 [7.71]	0.58 [8.61]	0.50 [7.85]	0.42 [6.65]	0.34 [5.65]	0.28 [4.78]
Quintile	Name	VW	0.30 [3.85]	0.31 [3.97]	0.30 [3.79]	0.28 [3.49]	0.23 [2.89]	0.22 [2.76]
Quintile	Cap	VW	0.25 [3.27]	0.26 [3.36]	0.25 [3.23]	0.21 [2.70]	0.18 [2.34]	0.16 [2.02]
Decile	NYSE	VW	0.30 [3.20]	0.27 [2.85]	0.24 [2.58]	0.21 [2.17]	0.20 [2.10]	0.17 [1.84]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.30 [3.98]	0.32 [4.18]	0.29 [3.85]	0.27 [3.56]	0.22 [2.92]	0.21 [2.75]
Quintile	NYSE	EW	0.34 [4.62]	0.41 [5.62]	0.33 [4.84]	0.29 [4.28]	0.18 [2.79]	0.15 [2.44]
Quintile	Name	VW	0.26 [3.38]	0.27 [3.53]	0.26 [3.33]	0.25 [3.17]	0.20 [2.65]	0.20 [2.58]
Quintile	Cap	VW	0.21 [2.80]	0.22 [2.92]	0.21 [2.77]	0.19 [2.48]	0.16 [2.12]	0.15 [1.94]
Decile	NYSE	VW	0.26 [2.77]	0.22 [2.40]	0.20 [2.14]	0.18 [1.92]	0.16 [1.78]	0.15 [1.65]

Table 3: Conditional sort on size and EDC

This table presents results for conditional double sorts on size and EDC. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on EDC. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high EDC and short stocks with low EDC. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	EDC Quintiles					EDC Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.45 [1.73]	0.70 [2.64]	0.83 [3.31]	0.93 [3.75]	1.03 [4.51]	0.61 [6.55]	0.67 [7.31]	0.61 [6.90]	0.55 [6.14]	0.42 [4.99]	0.38 [4.50]
	(2)	0.57 [2.50]	0.74 [3.03]	0.87 [3.69]	0.89 [3.91]	0.96 [4.46]	0.40 [4.50]	0.44 [4.96]	0.37 [4.24]	0.32 [3.63]	0.26 [3.02]	0.23 [2.64]
	(3)	0.52 [2.67]	0.62 [2.81]	0.83 [3.78]	0.81 [3.91]	0.93 [4.80]	0.41 [5.04]	0.42 [5.13]	0.38 [4.69]	0.35 [4.19]	0.29 [3.50]	0.27 [3.20]
	(4)	0.50 [2.77]	0.64 [3.15]	0.78 [3.82]	0.76 [4.02]	0.79 [4.38]	0.29 [3.72]	0.29 [3.65]	0.23 [3.00]	0.23 [2.86]	0.06 [0.79]	0.07 [0.92]
	(5)	0.48 [2.96]	0.49 [2.71]	0.52 [2.96]	0.52 [3.15]	0.71 [4.48]	0.23 [2.55]	0.23 [2.49]	0.23 [2.47]	0.20 [2.19]	0.19 [2.07]	0.18 [1.91]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	EDC Quintiles					EDC Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	361	360	359	359	353	31	33	37	29	28	
	(2)	105	103	104	104	103	56	55	56	55	54	
	(3)	76	77	75	76	76	97	96	96	100	99	
	(4)	64	64	64	64	64	200	205	209	211	215	
(5)	59	59	59	59	59	1436	1551	1565	1649	1888		

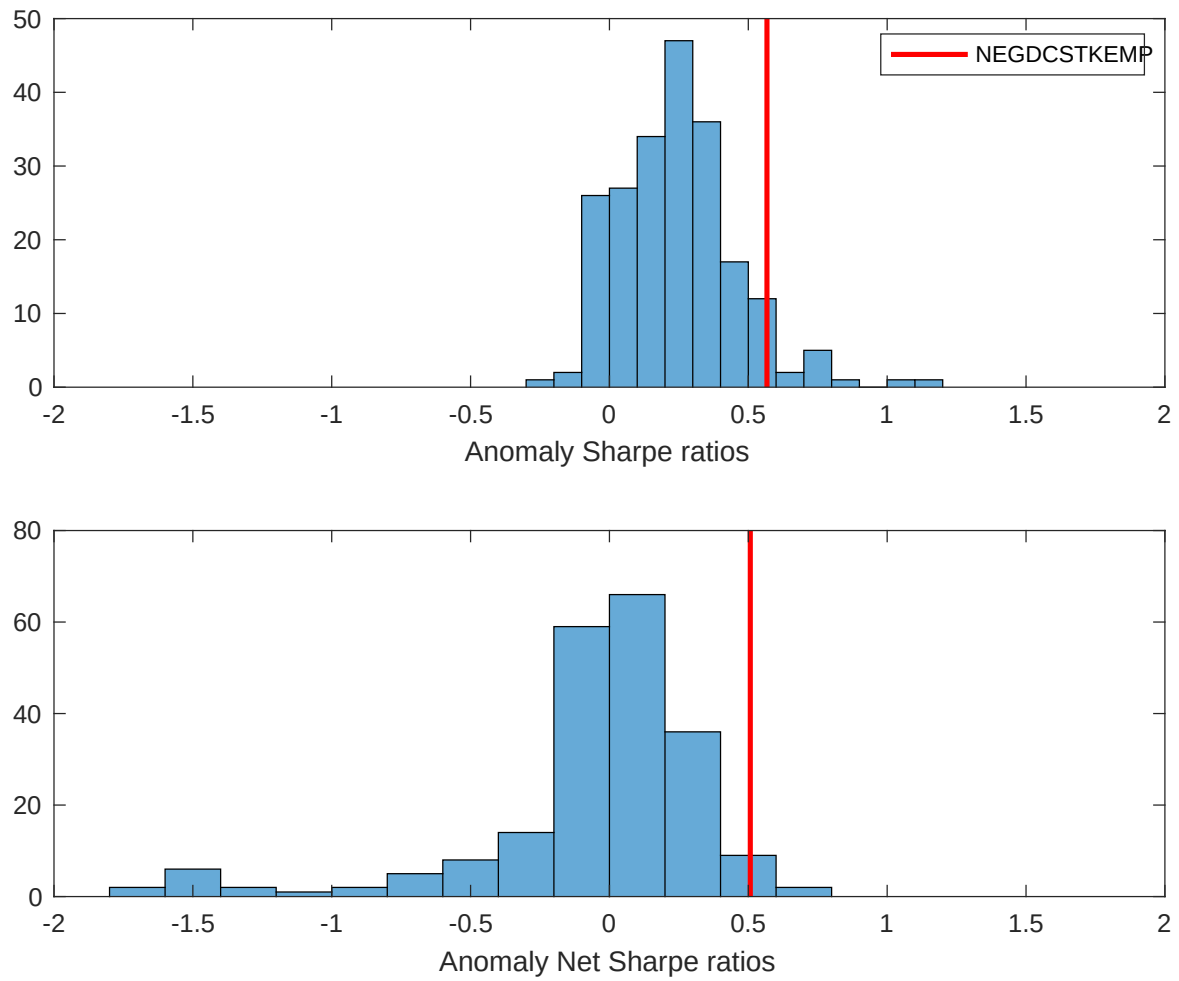


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the EDC with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

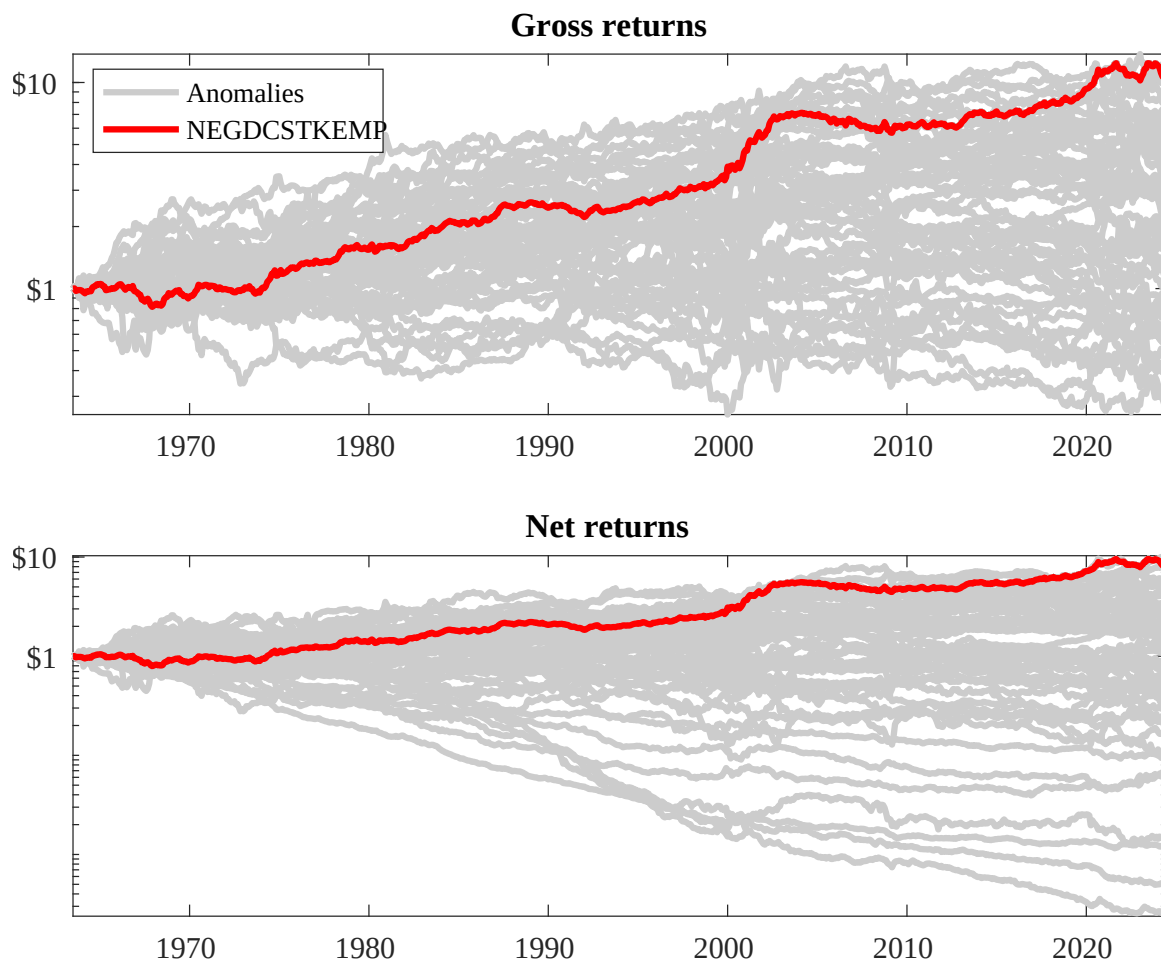
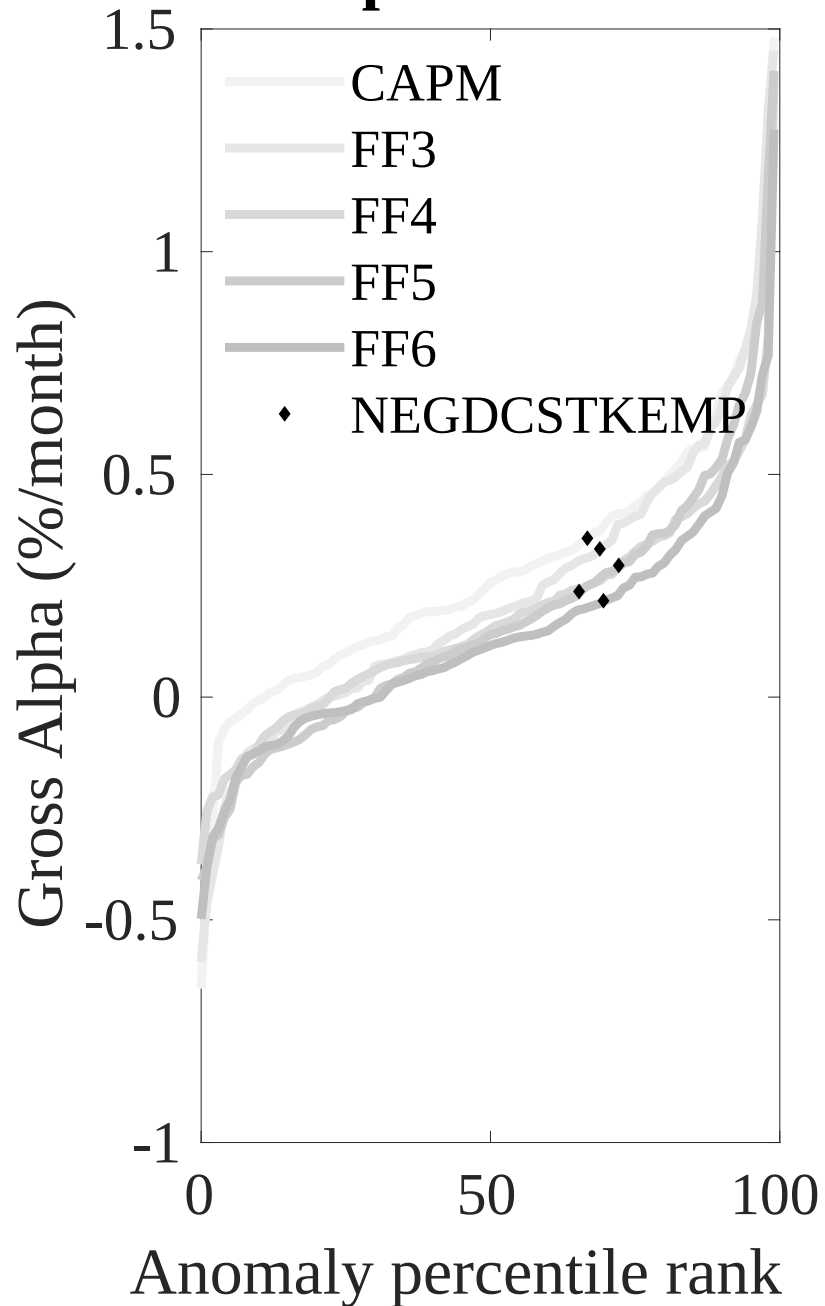


Figure 3: Dollar invested.

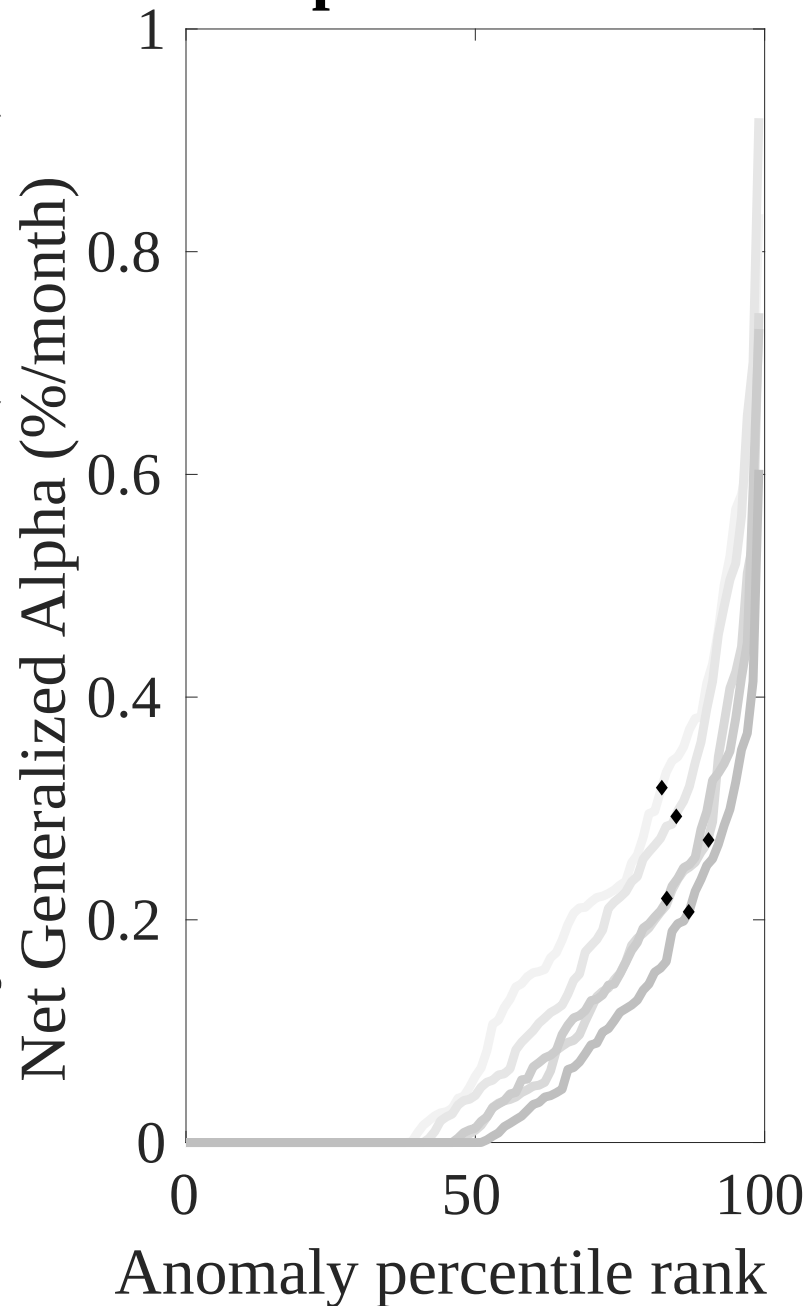
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the EDC trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



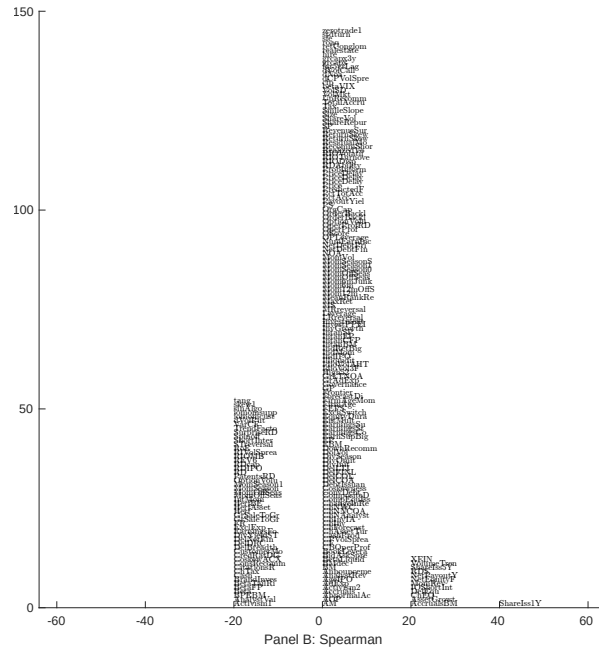
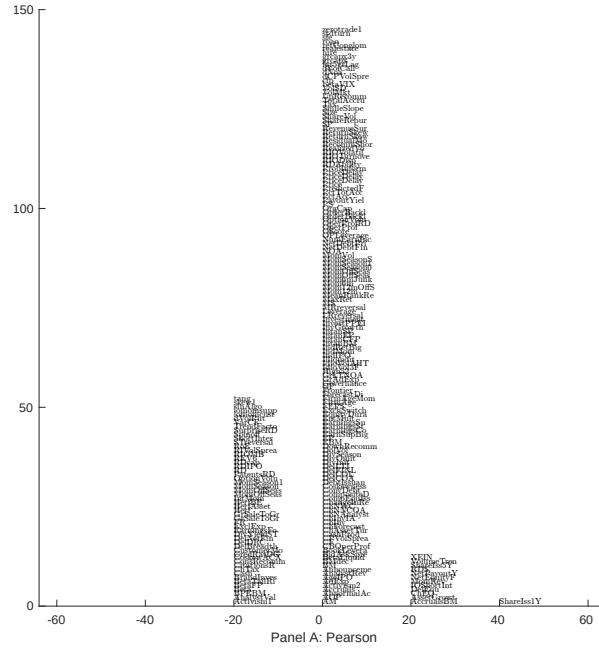


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with EDC. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

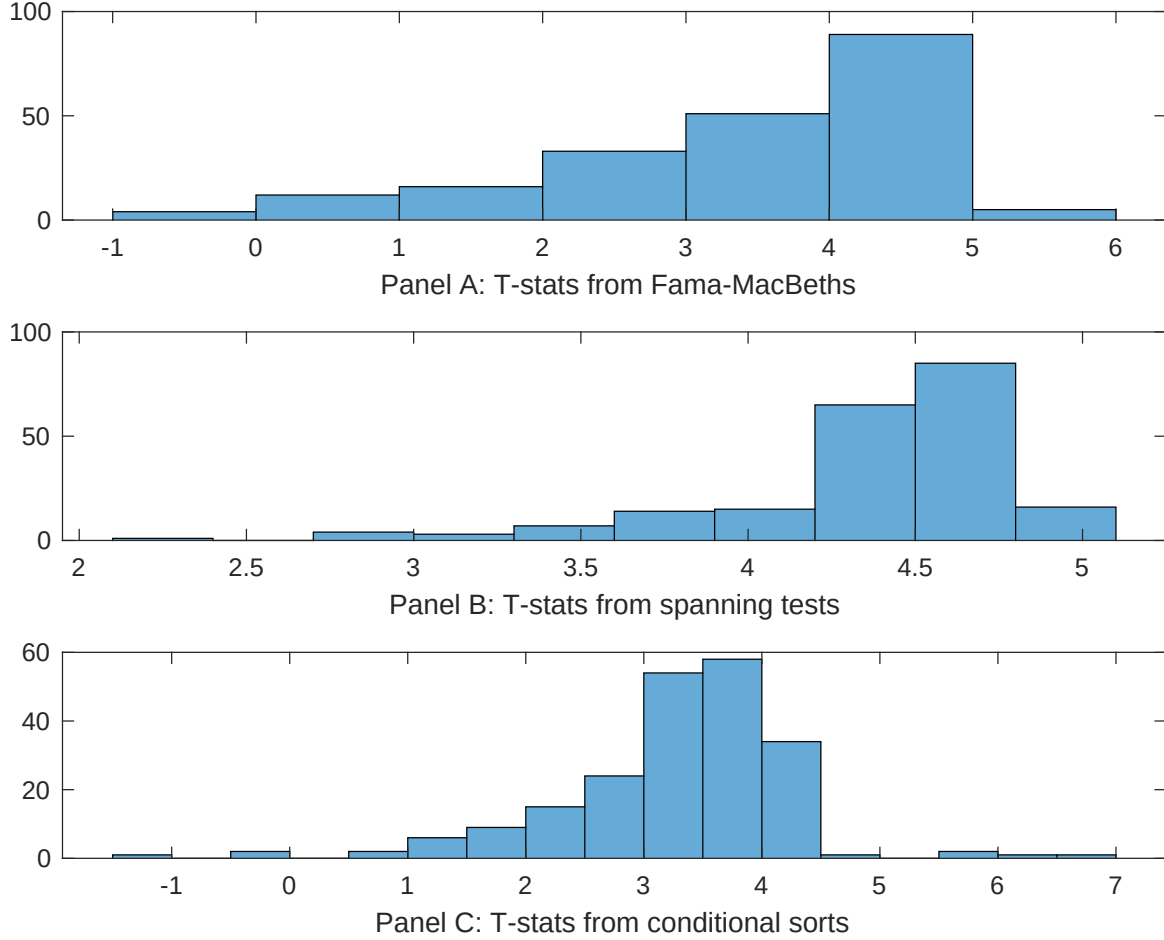


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of EDC conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{EDC} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{EDC}EDC_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{EDC,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on EDC. Stocks are finally grouped into five EDC portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted EDC trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on EDC. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{EDC}EDC_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Share issuance (1 year), Share issuance (5 year), Growth in book equity, Net Payout Yield, Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.13 [5.99]	0.13 [6.41]	0.17 [6.78]	0.12 [5.42]	0.12 [5.65]	0.13 [6.11]	0.11 [3.67]
EDC	0.47 [3.53]	0.44 [3.12]	0.51 [3.93]	0.46 [3.39]	0.53 [4.10]	0.46 [3.60]	0.31 [2.34]
Anomaly 1	0.14 [5.66]						0.73 [2.53]
Anomaly 2		0.27 [4.89]					0.23 [4.23]
Anomaly 3			0.39 [3.08]				-0.22 [-0.98]
Anomaly 4				0.20 [1.69]			0.98 [0.90]
Anomaly 5					0.12 [3.08]		0.67 [0.11]
Anomaly 6						0.92 [7.49]	0.64 [5.39]
# months	715	715	720	715	720	720	715
$\bar{R}^2(\%)$	0	1	0	1	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the EDC trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{EDC} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Share issuance (1 year), Share issuance (5 year), Growth in book equity, Net Payout Yield, Change in equity to assets, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.20 [2.63]	0.15 [1.93]	0.20 [2.67]	0.24 [3.17]	0.21 [2.78]	0.21 [2.71]	0.15 [2.12]
Anomaly 1	27.66 [7.29]						13.31 [2.91]
Anomaly 2		27.18 [6.75]					12.79 [2.82]
Anomaly 3			32.00 [7.95]				38.42 [6.60]
Anomaly 4				15.70 [5.32]			4.21 [1.32]
Anomaly 5					12.84 [3.36]		-11.62 [-2.17]
Anomaly 6						-0.25 [-0.05]	-16.29 [-3.25]
mkt	2.57 [1.44]	3.12 [1.73]	2.95 [1.66]	4.33 [2.32]	1.66 [0.91]	1.90 [1.03]	4.87 [2.73]
smb	7.70 [2.98]	8.42 [3.23]	5.54 [2.16]	9.14 [3.42]	6.45 [2.43]	6.64 [2.46]	8.94 [3.48]
hml	-8.11 [-2.35]	-5.19 [-1.49]	-11.32 [-3.27]	-13.45 [-3.63]	-9.07 [-2.53]	-7.32 [-2.05]	-11.11 [-3.14]
rmw	-6.65 [-1.64]	-1.41 [-0.37]	9.81 [2.83]	-0.81 [-0.21]	9.47 [2.64]	8.30 [2.30]	-4.98 [-1.18]
cma	16.12 [2.90]	20.11 [3.70]	2.33 [0.37]	20.05 [3.52]	20.40 [3.15]	33.63 [4.40]	9.59 [1.25]
umd	1.84 [1.04]	1.33 [0.74]	2.54 [1.45]	4.36 [2.41]	3.43 [1.88]	3.14 [1.71]	0.53 [0.30]
# months	728	728	732	728	732	732	728
$\bar{R}^2(\%)$	13	12	14	10	8	7	21

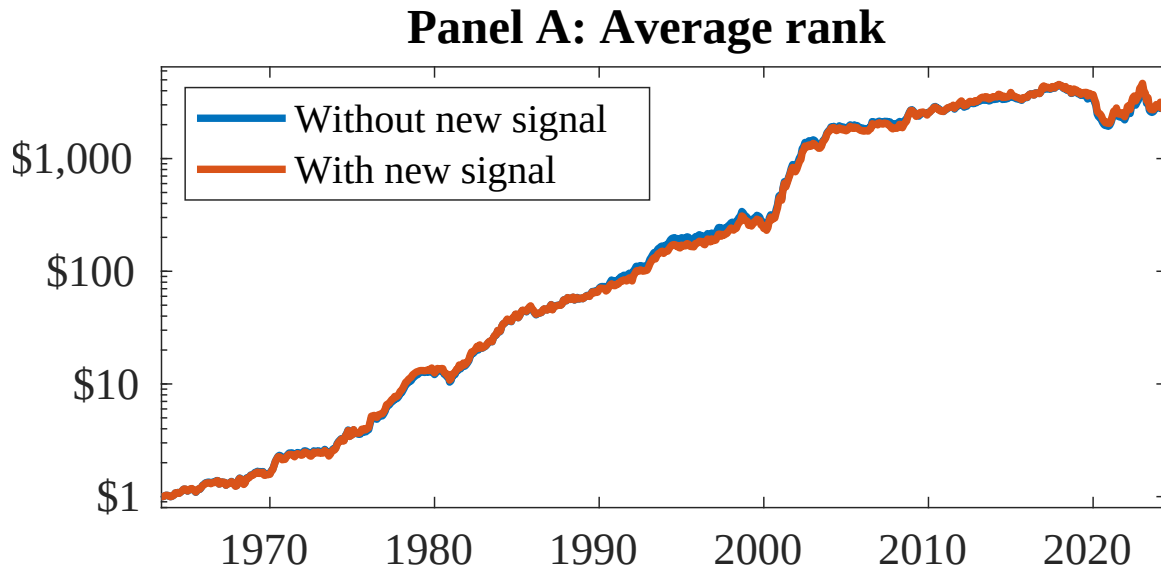


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as EDC. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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